GROUNDTRUTH ARCHITECTURE SUITE

Capturing the World Invariant

How to extract absolute environmental truths without imagining numbers or hallucinating rules.

ARCHITECTURE STANDARD
Zero Hallucination

THE CORE AXIOM OF GROUND TRUTH

A World Invariant is what the environment strictly enforces—independent of what an agent wishes to do. If the source text lacks concrete numbers, never invent them. Ground truth means extracting the exact mathematical boundaries, universal negations, and tripwires that exist in reality.

GROUND TRUTH IN ACTION: THREE DIFFERENT WORLDS

Literature / Fiction
e.g., Where the Mountain Meets the Moon

WORLD PHYSICS INVARIANT
"Nothing grows on Fruitless Mountain, and birds and animals do not rest there."

CHARACTER AXIOM
Minli has a habit of quick thinking and impulsive action ("Too well," sighed her mother).

Agent Boundary: Cannot invent farms on Fruitless Mountain; must route agent into a departure quest.

Enterprise IT Security
e.g., SOC2 Production Access Policy

CONSTITUTIONAL INVARIANT
Hardware FIDO2 WebAuthn is strictly required; legacy SMS OTP is permanently revoked.

TIME-BOUND CEILINGS
Direct DB write lease max TTL: 4 hours. Emergency Break-Glass token hard cap: 2 hours.

Agent Boundary: Auto-rejects peer approvals; triggers SOC2 alerts if break-glass exceeds 7,200s.

Industrial Physics & MPC
e.g., Model Predictive Control

PHYSICAL LAW INVARIANT
Conservation of mass; process bounds dwell in [u_min, u_max] under bounded disturbances.

RECEDING HORIZON INVARIANT
Solves finite horizon [t, t+T]; executes only 1st step u(t); shifts horizon forward.

Agent Boundary: If QP solver is infeasible, cannot guess an action—must trip emergency safe hold.

HOW TO EXTRACT INVARIANTS (THE 3 GOLDEN PRINCIPLES)

1

Never Hallucinate Constants
If Wikipedia defines a cost function algebraically as $J = \sum w(r-x)^2$, **do not fabricate a 42.0 kPa limit**. Ground truth means expressing algebraic constraints when numbers are unspecified.

2

Environment Physics ≠ Agent Strategy
Separate what the *world enforces* from what an actor *chooses*. Policies can be negotiated; world invariants (gravity, expiration timers, barren soil) never bend for any prompt.

3

Every Invariant Needs a Tripwire
An invariant is useless as passive prose. It must be wired into the execution DAG as a **circuit breaker**: if an observed state crosses the boundary, execution freezes instantly.

THE GROUNDTRUTH EXECUTION PIPELINE

